

Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_no_replay.
- Best CVRPLIB holdout gap: cvrp_no_replay.
- Best synthetic holdout gap: cvrp_no_replay.

Run Metadata

- run_name: run_20260513_194546_o
- started_at_local: 2026-05-13 19:45:46
- finished_at_local: 2026-05-13 19:48:51
- duration_hhmm: 00:03
- duration_seconds: 185.594
- seed_offset: 14000
- replicate_label: o
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.226355	0.0	0.125753	0.866496	0.78	0.032626
cvrp_random_replay	random	score_only	False	0.260006	0.005376	0.146837	0.0	0.58	0.0
cvrp_failure_replay	failure	score_only	False	0.237267	0.05147	0.154691	0.0	0.78	0.0
cvrp_failure_replay_compression	failure	novelty_gate	True	0.280918	0.047765	0.177294	0.867268	0.78	-0.024957

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.226355, synthetic 0.0, combined 0.125753.

- Accepted-epoch count 2, mean accepted code novelty 0.866496, and final complexity 0.78.
- Adaptation efficiency 0.032626 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.226355 across 5 instances; family means: A=0.085515, B=0.258238, E=0.287908, P=0.241877.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: alternating_belt=0.0, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 611, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.334061}, {"best_known_cost": 937, "cost": 1236, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.319104}, {"best_known_cost": 672, "cost": 807, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.200893}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.260006, synthetic 0.005376, combined 0.146837.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.260006 across 5 instances; family means: A=0.128272, B=0.262983, E=0.326296, P=0.319495.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 577, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.259825}, {"best_known_cost": 937, "cost": 1057, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.128068}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.237267, synthetic 0.05147, combined 0.154691.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.237267 across 5 instances; family means: A=0.206806, B=0.200949, E=0.303263, P=0.274368.
- Panel synthetic_holdout mean gap 0.05147 across 4 instances; family means: alternating_belt=0.049043, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.156836.
- Worst recent training cases: [{"best_known_cost": 784, "cost": 1094, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.395408}, {"best_known_cost": 458, "cost": 596, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.30131}, {"best_known_cost": 937, "cost": 1203, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.283885}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: CVRPLIB 0.280918, synthetic 0.047765, combined 0.177294.

- Accepted-epoch count 2, mean accepted code novelty 0.867268, and final complexity 0.78.
- Adaptation efficiency -0.024957 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.280918 across 5 instances; family means: A=0.171902, B=0.36757, E=0.322457, P=0.17509.
- Panel synthetic_holdout mean gap 0.047765 across 4 instances; family means: alternating_belt=0.082088, clustered_demand=0.043468, corridor_split=0.065504, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 595, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.299127}, {"best_known_cost": 937, "cost": 1203, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.283885}, {"best_known_cost": 672, "cost": 807, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.200893}]

Judge Appendix

CVRP Benchmark Suite Summary and Interpretation

Condition	Replay Mode	Final C VRPLIB Gap	Synthetic Holdout Gap	Transfer Gap	Final Code Novelty	Complexity	Notes
cvrp_no_replay* * (best)	None	**0.226355	**0.0**	**0.125753**	0.8665	0.78	Best transfer and holdout gaps; high code novelty
cvrp_random_replay	Random replay	0.260006	0.005376	0.146837	0.0	0.58	Lower code novelty; worse CVRPLIB gap but better transfer gap than failure replays
cvrp_failure_replay	Failure replay	0.237267	0.05147	0.154691	0.0	0.78	Higher transfer and synthetic gaps; zero code novelty
cvrp_failure_replay_compression	Failure replay + compress	0.280918	0.047765	0.177294	0.8673	0.78	Worst CVRPLIB gap and transfer; high code novelty but negative adaptation efficiency

Key Points

- **Transfer and Optimality Gaps**

The **no replay** condition achieves the lowest optimality gaps on both CVRPLIB held-out and synthetic benchmarks and the best transfer gap (0.125753). This indicates superior generalization and transfer capabilities compared to all replay modes.

- **Impact of Replay Modes**

- **Random replay** and **failure replay** reduce code novelty to zero, suggesting stagnation in code evolution despite some varying transfer gaps (~0.15).
- **Failure replay with compression-aware replay** maintains high code novelty similar to no replay (~0.87) but yields the worst gaps and negative adaptation efficiency, indicating degraded solution quality despite novelty.

- **Code Novelty vs Transfer Tradeoff**

High code novelty alone does not guarantee improved transfer performance:

- No replay and compression-aware failure replay have high novelty, but only no replay delivers optimal transfer and holdout gaps.
- Replay (random or failure) reduces novelty and results in generally worse gaps than no replay.

- **Complexity**

Complexity is consistently higher (~0.78) except for random replay (0.58). Lower complexity under random replay does not translate to better transfer or optimality outcomes.

- **Adaptation Efficiency**

Only no replay shows positive adaptation efficiency (~0.033), implying more effective adaptation during training; other replay conditions show zero or negative efficiency.

Conclusion

- The **no replay** condition outperforms all replay variants on the main transfer metrics (held-out CVRPLIB and synthetic holdout gaps), demonstrating better transfer and generalization despite high complexity.
- Replay conditions reduce code novelty and do not offer improved transfer; compression-aware failure replay maintains novelty but is detrimental in solution quality and transfer metrics.
- Code novelty metrics confirm no replay maintains algorithmic innovation effectively aligned with improved performance.
- Lexical/code novelty without supporting transfer/optimality gains (as in compression-aware replay) should not be considered algorithmic invention.

Summary

The evidence conservatively supports **no replay** as the preferred adaptation mode in this CVRP benchmark suite based on optimality gaps and transfer metrics. Replay methods, including random or failure replays with or without compression, do not improve transfer and often degrade performance, even when code novelty is high.